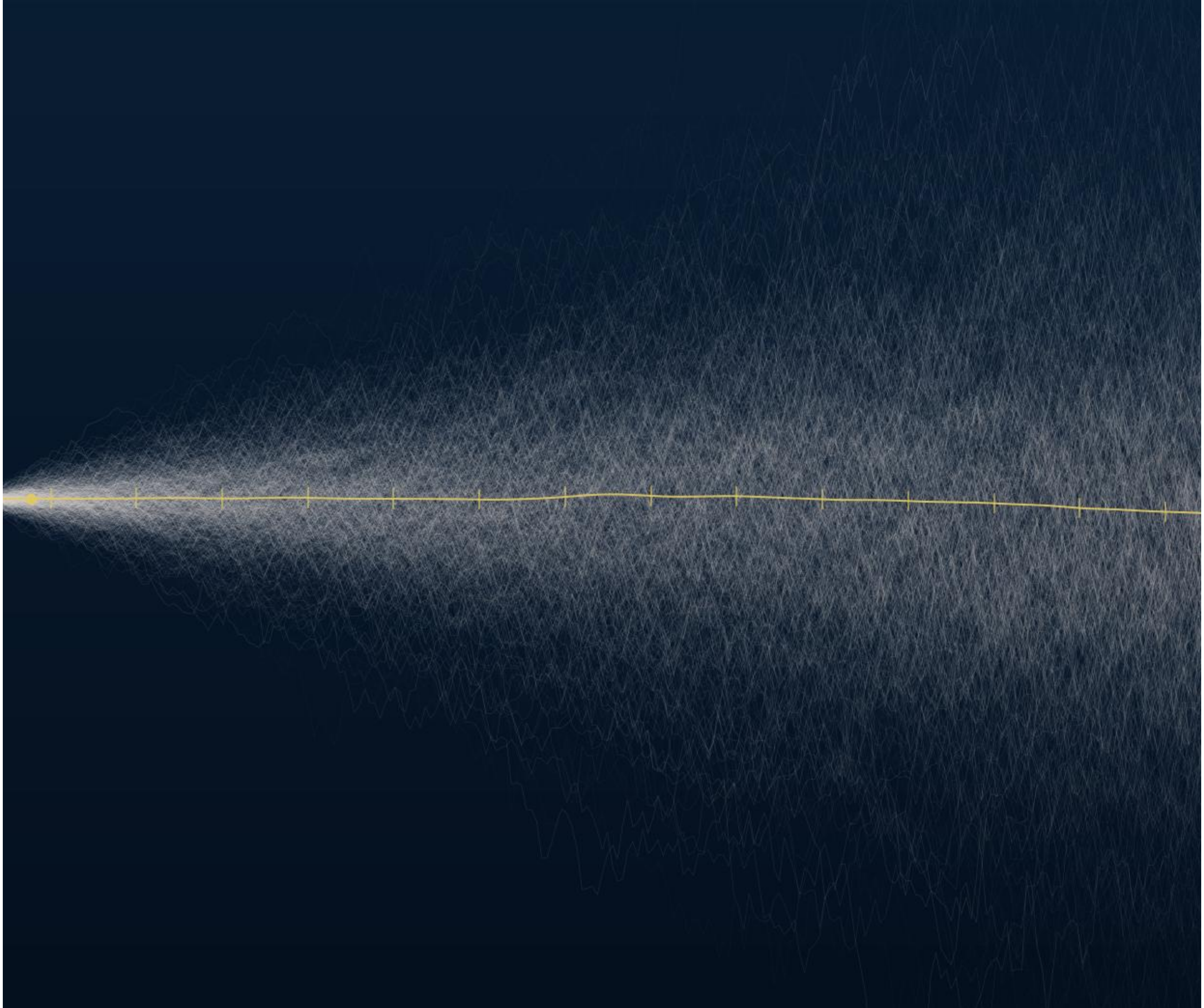
Australian Retirement Planning Tool

Quick Start Guide



Version 17.1

Contents

Welcome

This calculator helps you model your Australian retirement finances using sophisticated analysis tools including Monte Carlo simulation, historical stress testing, couple tracking, aged care modelling, and dynamic spending guardrails. This guide covers the core workflow and all major features.

The interface uses a **two-column layout**: inputs on the left, results and charts on the right. Sections are collapsible – click a section header to expand or collapse it.

A Note About Your Results

You can display your results in Real or Nominal dollars. Real dollars show you your results as purchasing power relative to either today's dollars, or to the first year of retirement (your choice). Nominal dollars shows you the actual amount that you would, for example, expect to see in your account in that year. Select your preferred output format using the three buttons in the top right of the screen. You can switch between them at any time.

Your First Simulation in 5 Steps

This section walks through each input field in the order you'll encounter it on screen, from top to bottom.

At the top of the inputs panel, a **dollar mode toggle** lets you enter spending, income, and super balances in either **today's dollars (2026 $)** or **retirement-year nominal dollars (Ret $)**. The calculator converts automatically using your CPI rate and years to retirement. Fields affected by the toggle show a small coloured badge in their label so you always know which denomination is active.

Step 1: Household Parameters

These are the first fields you'll see at the top of the left column, under the blue "Initial Situation" heading.

- **Base Annual Spending in Retirement:** How much you'll spend in your first year of retirement. Include groceries, utilities, insurance, entertainment, and regular travel. If retirement is several years away, switch to **2026 $** mode using the toggle at the top of the panel and enter your current spending — the calculator will inflate it to retirement-year dollars automatically. Typical range: $50,000–$100,000 for couples, $30,000–$60,000 for singles.
- **Sequencing Buffer:** A separate cash reserve to avoid selling investments during market downturns. Withdrawals follow: Cash → Buffer → Main Super. The return rate for this is defaulted to 3% but can be configured in the **Cash / Buffer Return Rate** field. Leave at $0 initially – this is an advanced strategy.
- **CPI / Inflation Rate:** Annual inflation rate used for projections. Affects spending growth, pension indexation, and real returns. The Australian long-term average of 2.5% is a sensible default.

Step 2: Your Super and Personal Details

Directly below Household Parameters, you'll see these fields.

Ages

- **Age Turning This Year:** The age you will, or did, turn this calendar year.
- **Retirement Age:** When you plan to stop working.
- **Sex:** Used to set the default Modelled Death Age from ABS life expectancy tables. Affects the simulation horizon in single mode.

- **Modelled Death Age:** The age the simulation runs to. Defaults to the ABS 2020–2022 life expectancy for your sex and current age, and updates automatically if you change either. You can override it manually. This drives the simulation horizon in single mode — the number of years simulated equals Modelled Death Age minus Retirement Age plus 1. A "Reset to ABS default" link appears if you have changed the value.
- **Variable death age in MC:** When enabled, each parametric Monte Carlo run samples a different death age from a normal distribution centred on your ABS life expectancy, reflecting genuine longevity uncertainty. Historical MC always uses fixed 35-year windows. See the Longevity & Outcome Analysis chart below for how to interpret the results.

Super and Pension

- **Super Balance at Retirement:** Your projected superannuation balance at the date of retirement. If retiring soon, use your current balance. If years away, use an online super projection calculator or your fund's estimate. You can express this figure in retirement-year dollars (default) or, if you switch to **2026 $** mode, in today's dollars — the toggle converts the denomination only. Either way, the figure must represent your expected balance at retirement, not your balance today.
- **Defined Benefit Pension / Superannuation Annuity (net annual):** Annual income from defined benefit pensions or lifetime annuities, after tax, as at the date of retirement. Examples: PSS, CSS, DFRDB, or commercial annuities. Use the estimate provided by your fund. The calculator indexes this to CPI automatically from Year 1 onward.

Step 3: Income During Retirement

These fields appear below the age selectors, still within the Initial Situation section. Start Year, Duration, Index to CPI and Employment Income will appear after a non-zero number is entered into Other Net Income.

Other Net Income

- **Other Net Income (optional, after tax):** Annual income from all non-super sources during retirement: part-time work, rental property, dividends, interest. This is counted in the Age Pension income test.
- **Start Year:** Which year of retirement this income begins (Year 1 = first year of retirement).
- **Duration:** How many years this income lasts. Leave blank for indefinite.
- **Index to CPI:** When enabled (default), the amount grows with inflation. When disabled, enter the fixed dollar amount you'll receive.
- **Employment income — Work Bonus eligible:** If this income is from wages or self-employment, tick this checkbox. The Work Bonus exempts up to $7,800/year per eligible partner (age 67+) from the Age Pension income test, with unused entitlement accumulating in a bank up to $11,800/person. Do not tick for rental income, dividends, or foreign pensions. The same checkbox appears on each Additional Income Stream.

Additional Income Streams

Click "+ Add Income Stream" to add multiple income sources, each with its own description, amount, start year, duration, and CPI indexing. Useful for modelling an overseas pension starting at 65, rental income stopping at 80, or part-time work for the first 5 years.

Age Pension Settings

Below the Other Net Income fields, you'll find the Age Pension configuration.

- **Age Pension Recipient Type:** Single or couple – affects maximum payment amounts and asset test thresholds.
- **Own Home:** Whether you own your home. Homeowners have lower asset test thresholds than non-homeowners.

- **Include Age Pension:** Check this to model Centrelink payments. The calculator estimates entitlements based on your assets, income, and homeowner status starting at age 67.
- **Assessable Assets Outside Super:** Financial assets outside super that Centrelink counts in the means test (bank accounts, shares, investment property equity, vehicles). Enter the estimated value at retirement. Your principal home is excluded if you're a homeowner. Any income from these assets should also be entered in Other Net Income above.

💡 **Tip:** *Include Age Pension – it's a significant safety net that becomes more important as super balances decline in later retirement.*

Step 4: Choose Your Scenario

Move to the right column. Under "Test Scenarios", select one of five analysis types:

Scenario	What It Does	Best For
Constant Return	Single return rate (0–10%) applied every year	Quick sanity checks, understanding baseline
Monte Carlo	500–2,000 random scenarios using your return/volatility assumptions	Probabilistic success rates
Historical MC	Monte Carlo using real historical return data — choose from AU Balanced (default), AU Equities, or US S&P 500 (98 years, 1928–2025)	Most realistic probabilistic analysis
Stress Tests	Predefined stress scenarios: crashes, bear markets, high inflation, extreme longevity, health shocks, worst case	Comprehensive stress testing suite
Historical Period	Replay actual market periods — US: GFC, COVID, 1929 Depression, Dot-com, Stagflation, 1982 Bull; AU: 1980 Bull, 1987 Crash, 1991–2024 Equities, 1990–2024 Balanced	Stress testing against real events

💡 **Tip:** *Start with Constant 5% for your first run to understand the basics, then move to Historical MC for realistic planning.*

Step 5: Review Your Results

- Charts update automatically for Constant Return and Historical scenarios.
- For Monte Carlo and Historical MC, click the "Run" button and wait 2–5 seconds.
- Results appear in the right column. Click to expand the Executive Summary panel at the top.

Understanding Your Results

Executive Summary

Key Metrics

- **Success/Failure:** Success means money lasted the full retirement period with balance ≥ $0. Failure means funds ran out early.
- **Starting Balance:** Your total super at retirement.
- **Ending Balance:** What remains at the end – positive means money left over, zero means it lasted exactly.
- **Years Lasted:** How many years your money survived.

Charts

Portfolio Balance

- Shows your total balance trajectory across retirement.
- Green = Main super, Orange = Sequencing buffer, Purple = Cash account (Constant Return, Stress Tests, and Historical period Scenarios only)
- Toggle “Real $” vs “Nominal $” at the top of the screen to see inflation effects.
- In Monte Carlo mode, percentile bands show the range of outcomes (P10/P25/P50/P75/P90).

Income vs Spending

- Green lines = Income sources (pension, Age Pension, other).
- Red lines = Spending (base + splurge + irregular expenses).
- Goal: Income covers as much spending as possible, with super filling the gap.

Annual Spending Breakdown

- Stacked bars showing where money comes from each year: Super (blue), Cash (yellow), Buffer (orange), Age Pension (dark green), Defined Benefit (medium green), Other Income (light green).
- Grey dashed line shows the minimum required super drawdown (not shown in Monte Carlo modes).

Longevity & Outcome Analysis Chart

When Monte Carlo or Historical Monte Carlo is active and results have been run, a Longevity & Outcome Analysis chart appears below the Executive Summary. It has two tabs:

- **Longevity scatter:** One dot per simulation run. Green = solvent at end, red = ran out of money. The X axis shows end age, Y axis shows final balance. Red dots clustered at older ages indicate longevity risk; red dots spread across all ages indicate the plan is fragile to poor returns regardless of lifespan.
- **Ruin age distribution:** A histogram of failed runs only, showing the age at which the portfolio hit zero. More useful than the success rate alone — failures clustered after 90 need a modest tweak; failures clustered before 80 require a fundamental plan change. An empty histogram means all runs succeeded.

What's a Good Result?

Rating	Indicators	Action
✅ **Green**	Success rate 85%+, ending balance > $0, money lasted the full modelled horizon	On track – consider whether you can enjoy more
⚠️ **Yellow**	Success rate 75–85%, balance gets low in late retirement	Consider reducing spending or delaying retirement
🔴 **Red**	Success rate <75%, runs out in many scenarios	Major adjustments needed

Advanced Features

The following collapsible sections appear below the Initial Situation panel in the left column. Each can be expanded by clicking its header.

Retirement Spending

The first collapsible section below the Initial Situation panel.

Additional Splurge Spending

- Extra spending for specific retirement years (e.g., $20,000/year for travel).
- **Start Age and Duration:** When the splurge begins and how many years it lasts.
- **Ramp-Down Years:** Optionally taper the splurge spending gradually rather than stopping abruptly.

Withdrawal Strategy

Spending Pattern

- **Constant Real:** Constant real spending throughout retirement – a more conservative assumption.
- **Declining (JPMorgan):** Based on JP Morgan research showing real spending naturally declines in later retirement (go-go, slow-go, no-go years). This is the default and generally more realistic.

Spending Modifiers

- **Forgo Inflation Adjustment After Loss Years**: When enabled, spending stays flat in nominal terms (no CPI uplift) in any year following a portfolio loss. This means your real purchasing power dips slightly in bad years rather than drawing down more capital. Most effective in Monte Carlo mode where loss years occur regularly.
- **Guardrails (Dynamic Spending):** Guardrails automatically adjust your spending based on portfolio performance using the Guyton-Klinger method.
    - **Upper Guardrail**: If portfolio grows beyond this threshold (e.g., 20% above plan), spending increases by the adjustment percentage.
    - **Lower Guardrail**: If portfolio drops below this threshold (e.g., 15% below plan), spending decreases.
    - **Adjustment:** How much spending changes when a guardrail is triggered (e.g., 10%).

💡 **Tip:** *Guardrails are only an indication of how adjusting your spending during poor market returns can make your super last longer. The outcome may not represent an acceptable spending profile in all cases.*

One-Off & Irregular Expenses

Models major lumpy expenses.

Use Stochastic Irregular Expenses

- Enabled by default – models realistic irregular costs for transport, housing, and medical expenses.
- Uses probability-based modelling with inflation adjustment and cost variation.
- Each Monte Carlo run gets a different expense path. Change the seed to see different random paths in single-scenario mode.

Manual One-Off Expenses

- Add specific large expenses at particular ages (e.g., new car at 70, home renovation at 68).
- Each has a description, age, and amount (in future dollars of the year it occurs).

💡 **Tip:** *Leave stochastic expenses enabled for the most realistic projections. They model things like car replacements, home maintenance, and medical costs that people often forget to budget for.*

Aged Care Modelling

Tick “Include Aged Care” to enable.

- **Modelling Approach:** Probabilistic (age-based probability curves, requires Monte Carlo) or Deterministic (care begins at a fixed age you specify).
- **Duration:** Length of stay. Australian average is approximately 3 years.
- **RAD (Refundable Accommodation Deposit):** Lump sum paid on entry, refunded on exit. Typical range $300k–$600k.
- **Annual Ongoing Costs:** Basic daily fee plus means-tested care fee. Typically $50k–$80k/year.
- **Couple Settings:** When couple tracking is active, aged care applies to the surviving partner after a death scenario is selected. Configure the person-at-home spending percentage and whether the partner dies in care.

Debts & Liabilities

Tick “Include Debt Repayment” to enable. Add debts carried into retirement, each with principal amount, interest rate, repayment period, and optional extra payments. Debt payments are unavoidable – they are not subject to guardrails and withdrawals come from the standard Cash → Buffer → Super waterfall.

Couple Tracking

When the **Age Pension Recipient Type** is set to “Couple”, a “**Track partners individually**” toggle appears in the Initial Situation section. Enabling it unlocks:

- Individual super balances and pension incomes for each partner.
- Different retirement ages for each partner.
- Reversionary pension rates (e.g., 67% of defined benefit continues to surviving partner).
- Individual partner charts showing separate super balance and pension trajectories.

Death Scenario Modelling

When couple tracking is enabled, a **Simulation Length (years)** field controls how long the scenario runs — independently of the partner death ages below, which only affect survivor spending and pension rules. A Death Scenario selector lets you model the financial impact of one partner dying. Three options:

- **Both Alive:** Both partners are alive for the duration of the simulation (i.e. no death modelling).
- **Partner 1 Dies:** At the Modelled Death Age, their super transfers to the survivor, pension reduces to the reversionary rate, spending drops to the single-person level (default 70% of couple spending), and Age Pension switches to single rates.
- **Partner 2 Dies:** Same as above but the other partner survives.

💡 **Tip:** *Use death scenarios to stress-test whether the surviving partner will be financially adequate alone.*

Scenario Manager (Save/Load)

Accessible via the **Save/Load** button at the top of the screen. Lets you:

- **Save:** Store your current inputs as a named scenario.
- **Load:** Restore a previously saved scenario.
- **Export/Import:** Download scenarios as JSON files to share or back up, and import them later.
- **Delete:** Remove saved scenarios you no longer need.

What-If Comparison

Save multiple scenarios and then use the What-If Comparison panel (in the results column) to compare them side by side. Useful for comparing strategies like retiring at 60 vs 62, different spending levels, with vs without part-time income, or different investment return assumptions.

DOCX Export

Click the **Export Word** button (top right of the screen) to download your simulation results as a formatted Word document, including your input parameters, key metrics, and summary data.

Sensitivity Analysis

Click the "**Sensitivity Analysis — Return vs Spending**" panel in the results column to expand it (only available in Constant Return mode). It displays a colour-coded grid showing your ending portfolio balance at the Modelled Death Age across seven return rates (your current setting ±3% in 1% steps) and five spending levels (80%, 90%, 100%, 110%, 120% of your base spending). Your current settings are highlighted with a blue border. All other inputs — Age Pension, income streams, debt, aged care — are included in every cell; only return and spending vary.

Green cells = survived with >$500k remaining. Yellow = survived with less. Orange = depleted in the final third of the horizon. Red = depleted in the first two-thirds. Look at how far the green zone extends from your blue cell — this shows how sensitive your plan is to your return and spending assumptions. Note that each cell uses constant annual returns (no volatility); use Monte Carlo for sequence-of-returns risk.

Using Monte Carlo (Recommended)

Parametric Monte Carlo

Uses your specified expected return and volatility to generate random scenarios.

1. Select “Monte Carlo” under Test Scenarios.
2. Set the number of simulations (1,000 recommended).
3. Set Expected Return and Volatility. Use one of the four Asset Allocation presets — Conservative (4.5%/6%), Balanced (6.5%/10%), Growth (7.5%/14%), or High Growth (9%/18%) — or choose Custom to enter your own values. Balanced is the default and a reasonable starting point for a typical industry super fund.
4. Click “Run Monte Carlo”.

If you change any inputs after running, the button will pulse orange reminding you to re-run.

Historical Monte Carlo

Uses real historical return data instead of synthetic random returns. Choose your data series from three options: **AU Balanced 60/40 (1990–2024, default)** — the closest proxy for a typical Australian industry super fund; **AU Equities XAOA (1980–2024)** — ASX All Ords accumulation index; or **US S&P 500 (1928–2025, Shiller/Ibbotson)** — 98 years including the Great Depression, stagflation, and multiple crashes. Three sampling methods:

- **Block Bootstrap (recommended):** Samples overlapping blocks of historical years, preserving multi-year correlations. Adjustable block size (3–10 years).
- **Shuffled Years:** Randomly reorders individual years. Maximum diversity but loses year-to-year correlations.
- **35-Year Blocks:** Uses complete consecutive 35-year periods from history. Approximately 62 unique scenarios.

Interpreting Monte Carlo Results

- **P10 (worst 10%):** Can you live with this outcome? This is the key planning number.
- **P50 (median):** The most likely outcome. Is this acceptable?
- **P90 (best 10%):** Don’t bank on this – it’s optimistic.
- **Success Rate:** Percentage of simulations where money lasted the full period.

When Variable death age is enabled, income and spending bands narrow and may terminate before the chart end — this reflects runs ending at their sampled death age, not portfolio failure.

Target Success Rates

- **90%+** – Excellent
- **85–90%** – Very good
- **75–85%** – Moderate (consider adjustments)
- **<75%** – Needs work

Stress Tests

The Stress Tests mode runs predefined scenarios designed to probe specific vulnerabilities.

Test	Name	Description
A1	Base Case	5% return, 2.5% CPI – baseline reference
A2	Low Returns	3.5% return – structural low-growth test
B1	Crash	Immediate -25%/-15% crash then 5% recovery
B2	Bear Market	10 years of zero returns, then 5%
B3	High Volatility	Wild swings averaging 5% – tests sequence risk
C1	High Inflation	5% CPI entire period – tests purchasing power erosion
D1	Extreme Longevity	Survival to age 105 (45-year retirement)
G1	Health Shock	$30k/year additional medical from age 75
H1	Worst Case	Market crash + high inflation + health costs combined

Click on a formal test to see how your plan performed under that specific stress scenario in the graphs below. A robust plan should survive A1, A2, and at least some of the B-series tests.

Common Adjustments

If Running Out of Money

Option 1: Reduce Spending

- Lower base spending by 10–20%.
- Reduce or eliminate splurge spending.
- Cut one-off expenses.

Option 2: Delay Retirement

- Working 1–2 more years gives super more time to grow.
- Also reduces the number of retirement years to fund.

Option 3: Add Part-Time Income

- Use “Other Net Income” to add $10–20k/year for the first 5–10 years.
- Can significantly improve outcomes.

Option 4: Enable Guardrails

- Dynamic spending guardrails automatically reduce spending during poor market periods.
- Models a realistic adaptive spending strategy.

If Too Much Left Over

- Increase base spending.
- Add or increase splurge spending for travel and hobbies.
- Retire earlier.
- Or do nothing – a buffer provides peace of mind and legacy.

Pro Tips

1. Start Conservative

- Use realistic spending estimates – most people underestimate costs.
- 5% nominal return (roughly 2.5% real after inflation) is a reasonable starting point for a conservative portfolio (use 7% for a more equity-heavy allocation).
- Build in a sequencing buffer if you're concerned about early retirement market crashes.

2. Test Multiple Scenarios

- Run Constant 4%, 5%, 6% to see the range.
- Use Historical MC for the most grounded probabilistic view.
- Run formal tests B1 (Crash) and B2 (Bear Market) to stress test early retirement vulnerability.

3. Use Real Dollars View

- Select Real $ to see constant purchasing power. You have two options – the first button shows purchasing power in todays dollars; the second option shows purchasing power in the first year of retirement dollars
- Makes it much easier to compare across years – your brain thinks in constant purchasing power, not inflated future amounts.

4. Don't Obsess Over Precision

- This is planning, not prediction.
- Focus on broad patterns: Does the plan survive stress tests? Is the success rate adequate?
- Revisit annually and adjust as circumstances change.

5. Age Pension Matters

- Include it – it's a significant safety net.
- The calculator estimates it based on assets and income tests.
- It becomes more important as super balances decline in later retirement.

6. Use the Scenario Manager

- Save your base case before experimenting.
- Export scenarios as JSON files for backup.
- Use What-If Comparison to evaluate alternatives side by side.

Checklist for Your First Run

5-Minute Setup

- Enter base annual spending, sequencing buffer, and CPI rate.
- Enter super balance and defined benefit pension income.
- Set current age and retirement age.
- Add any other net income (part-time work, rental, etc.).
- Set single or couple and homeowner status.
- Enable Age Pension.
- Choose scenario: Start with Constant 5%.
- Review results – did it succeed?

15-Minute Analysis

- Run Historical Monte Carlo (1,000 simulations).
- Check success rate and percentile bands.
- Run Formal Tests B1 and B2.
- Adjust if needed and save your scenario.

Deep Dive

- Enable couple tracking and model death scenarios.
- Configure retirement spending pattern and splurge years.
- Add irregular expenses and aged care costs.
- Configure spending guardrails.
- Use What-If Comparison across multiple strategies.

This calculator is a planning tool, not financial advice. Use it to test strategies and build confidence in your retirement plan. Revisit regularly as circumstances change.